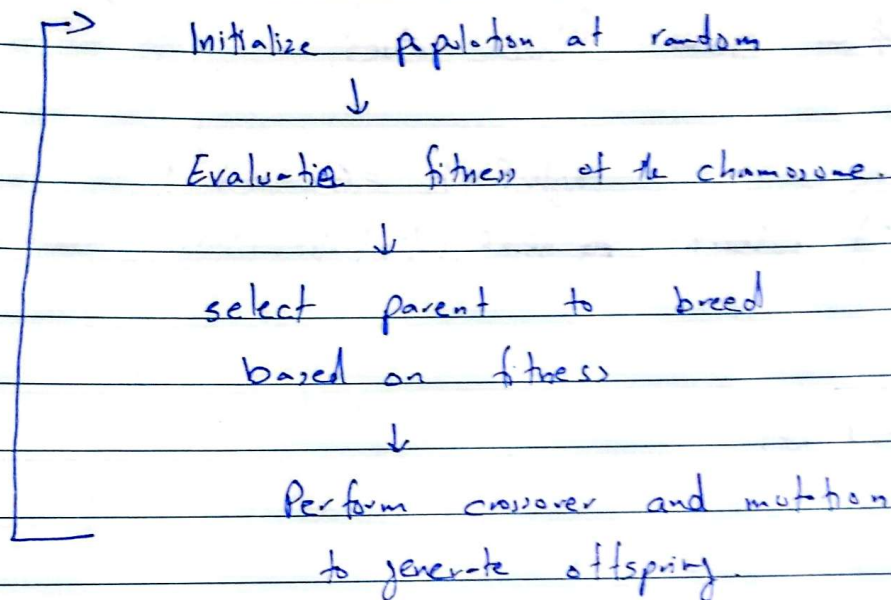


Genetic Algorithms



Selection Operators.

Tournament selection

- Multiple tournaments are conducted among randomly chosen individuals.
- The winner of each is selected for next gen.

Roulette wheel selection

- selected based on the fitness value
- higher fitness = greater chance of selection

Rank-based selection

- Assign probability to be selected based on their ^{rank} ~~fitness~~ rather than raw fitness values.

$$P_i = \frac{2 \times \text{Rank}}{N \times (N+1)} \quad \text{higher Rank} = 6 \text{ not } 1.$$

where $N =$ no. of individuals.

or

$$P_i = \frac{\text{rank of } i}{\text{Sum of Ranks}}$$

Random selection

Select randomly from the population.

Generation Gap

- between 0 or 1
- Represent fraction of population that gets replaced.
- for $G.P = 1$, all new offsprings
- for $G.P = 0.4$, $0.4 \times P$ are replaced while $1 - 0.4 \times P$ are preserved.

Elitism selection of k individuals

- Ensure a fraction of best individuals remain.

Crossover Operator

- single point

$$\begin{array}{c|c} 10 & 11 \\ \hline 11 & 00 \end{array} \rightarrow \begin{array}{c} 1000 \\ \text{or} \\ 1111 \end{array}$$

- Two point

$$\begin{array}{c|c|c} 10 & 11 & 00 \\ \hline 11 & 00 & 01 \end{array} \rightarrow \begin{array}{c} 100000 \\ \text{or} \\ 111100 \end{array}$$

Mutation

- Bit flip

$$0110 \rightarrow 0100$$

- Random resetting

$$0111 \rightarrow 1111$$

- Swap

$$1234 \rightarrow 1324$$



- Scramble

1 2 4 5 6 7 → 1 6 4 5 2 7

- Inversion

1 2 3 4 5 6 → 1 5 4 3 2 6

- Boundary replaces.

3.2 4.1 1.5 5.8 → 3 4 2 6

Termination

After reproduction, we may terminate if

- fitness threshold
- Generational limit.